

Maggie Ma

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TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, TypeScript, SQL, Verilog, HTML, GDScript

Frameworks & AI: React, Next.js, Tailwind CSS, OpenHex, Claude API, ElevenLabs, AI Agents

Robotics & Embedded: ROS, AirSim, Unreal Engine, ESP32, STM32, Arduino, Sensor Integration

Tools & Platforms: Git, Sequel Ace, SQLite, Arduino IDE, Excel, KiCad, Fusion 360, Blender, Godot

EXPERIENCE

UAV Research Intern — Tongji University | *ROS, Unreal Engine, UAV Systems* May 2026 – Aug. 2026

- Second author of a manuscript on **virtual physical fusion testing**, covering UAV closed loop validation and **LLM** scenario configuration; currently under review at the **China Journal of Highway and Transport**.
- Developed a **virtual-physical UAV testing platform** in **AirSim/Unreal Engine**, building repeatable low altitude scenarios with obstacles, virtual UAVs, and collision conditions for obstacle avoidance evaluation.
- Integrated **ROS control** with the **Nokov motion capture system**, using real time position and velocity feedback to support closed loop physical UAV testing and validate virtual physical interaction in a scaled environment.

EMG Fabric — Biotron Design Team | *ESP32, Arduino IDE, C++* Sep. 2025 – Dec. 2025

- Designed and refined low noise **EMG acquisition circuits**, optimizing wiring and sensor connections to improve signal integrity, measurement accuracy, and real time reliability.
- Integrated and debugged **ESP32-based hardware-software workflows**, tested **OpenEMG** and **MyoWare** sensors, and analyzed recorded EMG data using **SQLite** and **Excel**.

PROJECTS

Tongzhou — Alumni Association Mini Program | *React, OpenHex, SQL, SourceTree* Jul. 2026 – Aug. 2026

- Developed an **internal mini program** for an alumni association to provide alumni information, services, and event support in one platform.
- Used **Sequel Ace** and **SQL** to clean, filter, and consolidate alumni records, transforming raw survey and membership data into structured information for the platform.
- Trained and refined **OpenHex agents** using processed alumni data, sample queries, and iterative testing, enabling natural language **record lookup**, **information summarization**, and alumni service recommendations.
- Used **Docker** for local backend setup, **SourceTree** for Git branch management, and **CC Switch** to manage AI coding tool configurations during development.

Eight For Long — Room Escape Game | *Godot, GDScript, Game Design* Jul. 2026

- Designed and shipped a **room escape game** in 96 hours for **GMTK Game Jam 2026**, ranking in the **top 5% for Narrative** among **10.5k entries**, with an eight-minute countdown serving as the core resource for puzzles, interactions, and player choices.
- Developed **minigames and interactive puzzles** in Godot and created original art assets for the underwater environment and gameplay sequences.
- Released the game on **itch.io** and produced a **gameplay walkthrough video** demonstrating the full player experience.

Sprout | *TypeScript, Next.js, React, Tailwind CSS, ElevenLabs, Claude API* Mar. 2026

- Placed **3rd** in the University of Waterloo AI Data Science Hackathon.
- Built a **voice-first English-Rohingya learning platform** for Rohingya newcomers and refugees with low literacy, emphasizing accessibility through spoken interaction and guided UI flows.
- Developed core product features including **speaking assessment**, **conversation practice**, **AI-assisted vocabulary generation**, and user progress tracking.
- Integrated **ElevenLabs** and **Claude API** to support real time voice interaction and adaptive AI generated learning feedback.

EDUCATION

University of Waterloo | *BASc in Computer Engineering (Co-op)*

Sep. 2025 – Apr. 2030

- Waterloo, Ontario, Canada